Junsu Kim

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Research Interests

Computer Architecture, Systems for ML & ML for Systems

Education

University of British Columbia, Vancouver, BC, Canada Sep. 2025 - Current
Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Prashant Nair)
Cumulative GPA: 4.0/4.0

Korea University, Seoul, Korea Sep. 2023 - Aug. 2025
M.S. in Electrical Engineering (Advisor: Prof. Yunho Oh)
Cumulative GPA: 4.0/4.0

Hanyang University, Seoul, Korea Mar. 2014 - Feb. 2021
B.S. in Electronic Engineering (Advisor: Prof. Ki-Seok Chung)
Cumulative GPA: 3.81/4.0 (Graduating with Honors - Summa Cum Laude)

Publications

Conference [C] and Journal [J] Papers

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, and Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh "Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture" The IEEE Computer Architecture Letters (CAL), 2025.

[C5] **Junsu Kim**, and Suhyun Kim, "Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning", The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[C4] Seondeok Kim*, Sangun Choi*, Jaebeom Jeon, **Junsu Kim**, Minseong Gil, Jaehyeok Ryu, and Yunho Oh, "Kubism: Disassembling and Reassembling K-Means Clustering for Mobile Heterogeneous Platforms", The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[C3] Dongwon Yang, Jaebeom Jeon, Minseong Gil, **Junsu Kim**, Seondeok Kim, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "SSFFT: Energy-Efficient Selective Scaling for Fast Fourier Transform in Embedded GPUs", The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[J1] Minseong Gil, Jaebeom Jeon, **Junsu Kim**, Sangun Choi, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs", The IEEE Embedded Systems Letters (ESL), 2024.

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh. "VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing". The 53rd International Conference on Parallel Processing (ICPP), 2024

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung. "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer". The 39th IEEE International Conference on Computer Design (ICCD), 2021

Preprints (Project Names Only)

[P1] Co-author, "Hardware Supports for Enabling Arbitrary Numeric Format on GPUs"

Work Experience

University of British Columbia, Vancouver, BC, Canada Sep. 2025 - Current
Research Assistant at Systems and Architectures (STAR) Lab Advisor: Prof. Prashant Nair

Korea University, Seoul, Korea Sep. 2023 - Aug. 2025
Research Assistant at Computer Architecture and System Software Lab (ComSys) Advisor: Prof. Yunho Oh

Korea Institute of Science and Technology, Seoul, Korea May. 2022 - Aug. 2023
Research Assistant at Korea Data Science Team (KDST) Supervisor: Dr. Suhyun Kim

Hanyang University, Seoul, Korea Dec. 2019 - Mar. 2020, Aug. 2020 - Nov. 2020
Research Assistant at Embedded System on Chip Laboratory (ESOC Lab) Advisor: Prof. Ki-Seok Chung
Research Assistant at Computer Architecture and System SW Lab (CASS Lab) Advisor: Prof. Yongjun Park

Teaching Experience

University of British Columbia, Vancouver, BC, Canada — Fall 2025
Teaching Assistant for Computer Architecture (CPEN 411)

Korea University, Seoul, Korea — Spring 2024, Fall 2024
Teaching Assistant for Computer Architecture

School for the Blind, Chuncheon, Korea — Mar. 2017 - Feb. 2019
Assistant Teacher (Alternative Military Service)

Honors and Awards

Faculty of Applied Science Graduate Award, University of British Columbia (UBC) → $7200 CAD — Nov. 2025

AAAI Student Travel Grant → ∼ $1600 USD — Mar. 2025

Korea University Graduate School Scholarship → Half Tuition (∼ $10000 CAD) — Spring 2024, Fall 2024

Korea University Graduate School Scholarship for Outstanding New Student → Half Tuition (∼ $5000 CAD) — Fall 2023

Hanyang University Scholarship → Full Tuition (∼ $20000 CAD) — 2014 - 2019

Research Projects

Accuracy-preserving Tiered Key-Value (KV) Cache Management for Large Language Model (LLM)
Advisor: Prof. Prashant Nair, University of British Columbia — Sep. 2025 - Current

- Proposed a tiered KV cache management framework that leverages a full KV cache stored in host DRAM and a compressed KV cache residing in GPU HBM
- Proposed reforwarding that rejects a new output token generated by the compressed KV cache and generates a new one with full KV cache depending on output token confidence
- Proposed KV cache recomputation that recomputes KV cache generated by the compressed KV cache during reforwarding
- Reduced the KV cache size by 60% while achieving an average speedup of 65.4% on LongWriter (Long-generation) and Longbench-v1 (Long-context), with less than 1% accuracy loss.
- Contributions: first-author, motivation study, idea, implementation

Memory Oversubscription-aware Tensor Migration Scheduling for GPU Unified Storage Architecture [CAL]
Advisor: Prof. Yunho Oh, Korea University — Feb. 2024 - Sep. 2024

- Analyzed page faults due to memory oversubscription stalled AI workloads when expanding GPU memory with SSD using UVM
- Proposed a tensor migration scheduling algorithm considering GPU memory oversubscription for GPU unified storage architecture
- Achieved the averaged speedup by 12.9% compared to G10, which was presented at MICRO 2023
- Contributions: 1st author, motivation study, idea, implementation, experiment, paper write-up

Hardware Supports for Enabling Arbitrary Numeric Format on GPUs
Advisor: Prof. Yunho Oh, Korea University — May. 2024 - Nov. 2024

- Observed GPU supports a limited set of numeric formats, wasting register files when processing arbitrary numeric formats
- Employed bitslice representation, which transposes the data elements, packing arbitrary numeric formats without register wastage.
- Proposed Bitslice Vector multiplier and adder, constructing a tree structure to replace a multiplication-adder tree in a Tensor core
- Contributions: co-author, motivation study, idea, implementation, paper write-up

A Behavioral Analysis of CXL Memory Systems
Advisor: Prof. Yunho Oh, Korea University — Sep. 2023 - Sep. 2024
Collaborator: SK hynix

- Observed the behavior of a real CXL-based system on datacenter and AI workloads in the CXL-based platform
- Analyzed how the different promotion and demotion methods for CXL devices affected the performance of the workloads
- Presented performance modeling for datacenter workloads using different system factors (e.g., memory bandwidth, memory latency)
- Contributions: co-author, experiment, analysis, paper write-up

Salient Frequency-aware Exemplar Compression for Online Continual Learning [AAAI'25]
Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology — Jan. 2023 - Nov. 2023

- Observed exemplar compression methods occupied limited GPU resources during online continual learning
- Proposed a computationally efficient compression algorithm using salient frequency
- Proposed a buffer management scheme to alleviate harmful effects from the compression artifacts remaining in the buffer
- Contributions: 1st author, motivation study, idea, implementation, paper write-up

Skills

C/C++, Python, Tensorflow, Pytorch, Git, Verilog, Shell script